



odd | odd | swaps

Even swaps



































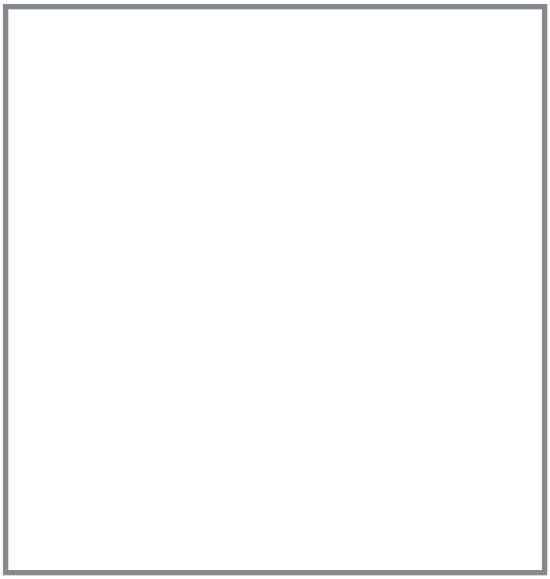


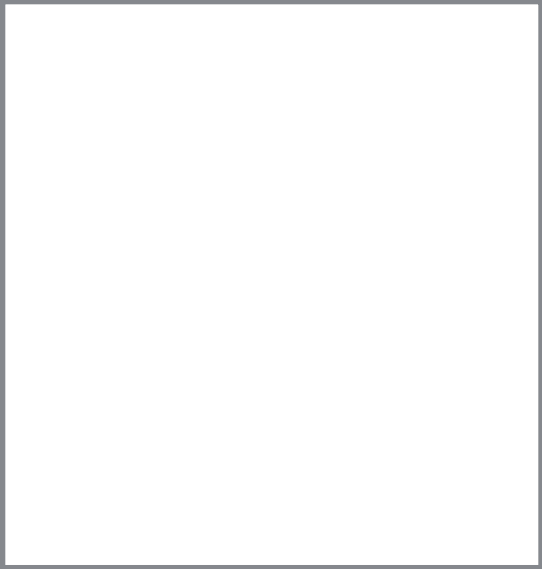












Example















































































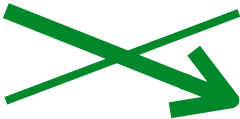


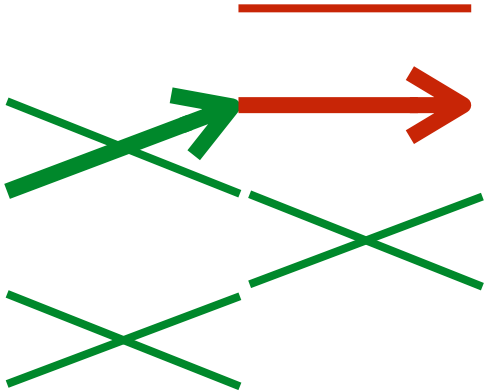
















Deterministically alternate between odd and even swaps

2

3

NON-REVERSIBLEPT (OKABETAL, 2001)

$$\mathbf{K}_{\text{comm},t} = \mathbf{K}_{\text{even}} 1(t \equiv 0 \bmod 2) + \mathbf{K}_{\text{odd}} 1(t \equiv 1 \bmod 2)$$

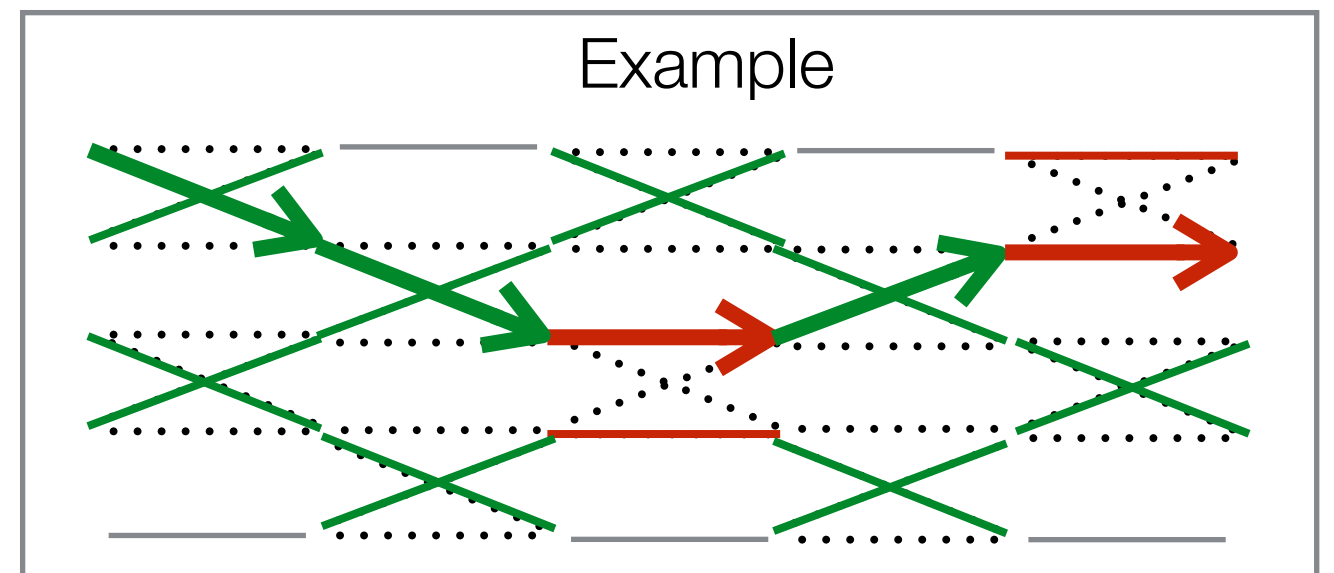
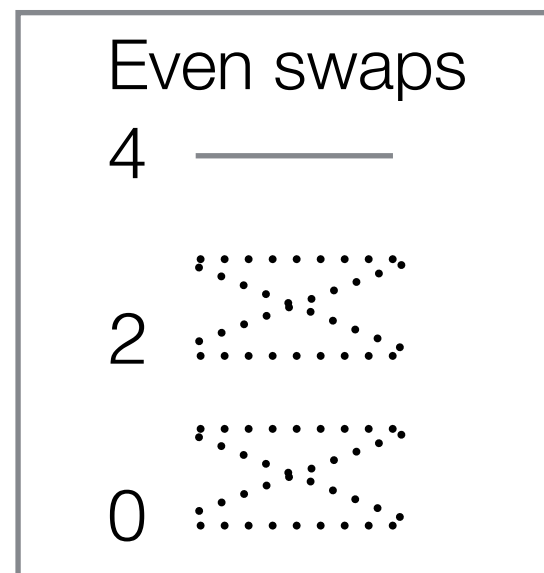
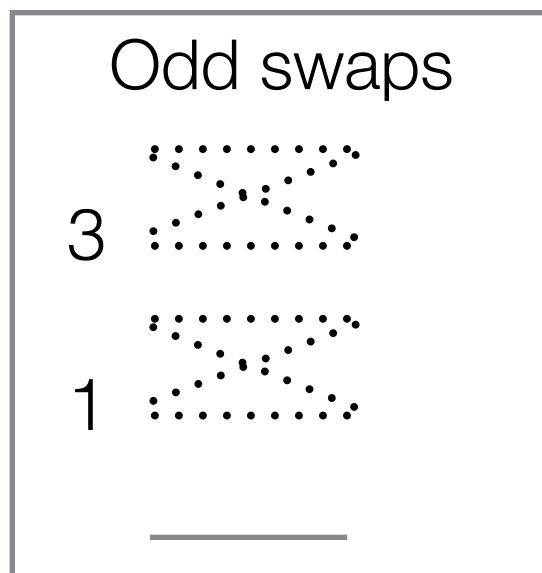
$$= \prod_{n \equiv t \pmod{2}} K^{(n, n+1)}$$

NON-REVERSIBLE PT (OKABE ET AL, 2001)

- Deterministically alternate between **Odd** and **Even** swaps

$$\mathbf{K}_{\text{comm},t} = \mathbf{K}_{\text{even}} 1(t \equiv 0 \pmod{2}) + \mathbf{K}_{\text{odd}} 1(t \equiv 1 \pmod{2})$$

$$= \prod_{n \equiv t \pmod{2}} \mathbf{K}^{(n,n+1)}$$



NON-REVERSIBLE PT (OKABE ET AL, 2001)

- Deterministically alternate between **Odd** and **Even** swaps

$$\mathbf{K}_{\text{comm},t} = \mathbf{K}_{\text{even}} 1(t \equiv 0 \pmod{2}) + \mathbf{K}_{\text{odd}} 1(t \equiv 1 \pmod{2})$$

$$= \prod_{n \equiv t \pmod{2}} \mathbf{K}^{(n,n+1)}$$